RPA analysis of the instability.

Continue $i\omega_n \to \omega + i\delta$ to obtain retarded GF

$$\Pi^{(0)}_{\uparrow\downarrow}(\bar{q}, i\omega_n \to \omega + i\delta) = \int \frac{d\bar{k}}{(2\pi)^3} \frac{f(\xi_\uparrow(k)) - f(\xi_\downarrow(k+q))}{\omega + i\delta + \xi_\uparrow(k) - \xi_\downarrow(k+q)}$$

But $\xi_\uparrow(k) - \xi_\downarrow(k+q) = \varepsilon(k) - \frac{\Delta}{2} - \left(\varepsilon(k+q) + \frac{\Delta}{2}\right) =$

$= \varepsilon(k) - \varepsilon(k+q) - \Delta \quad , \quad \varepsilon(k) = \frac{k^2}{2m}$

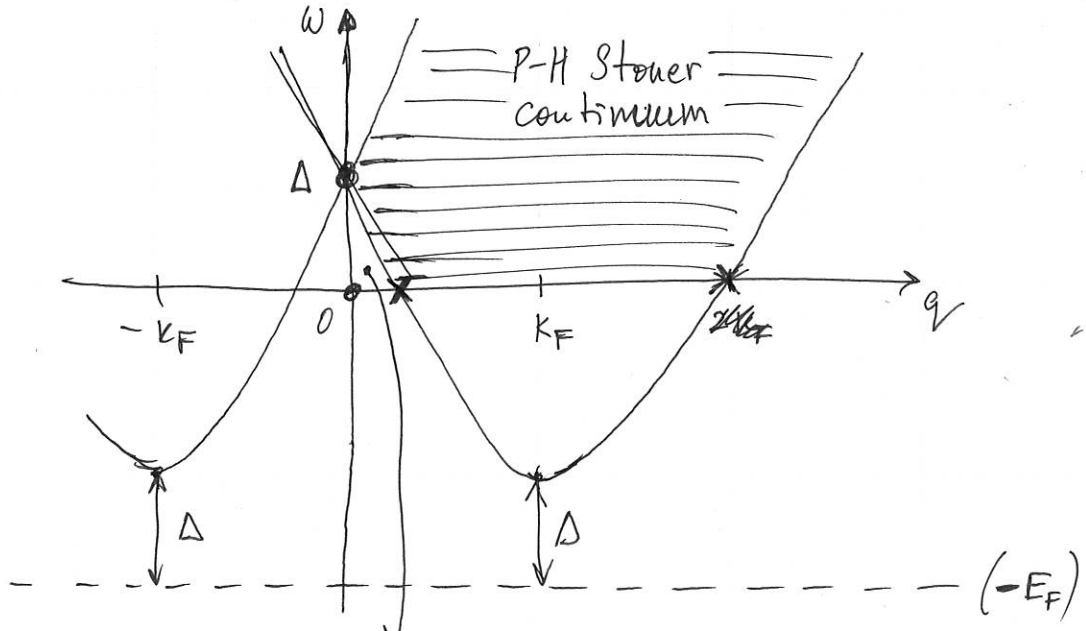
Re-write $\Pi^{(0)}_{\uparrow\downarrow} = \int \frac{dk}{(2\pi)^3} \left\{ \frac{f(\xi_\uparrow(k))}{\omega + i\delta + \xi_\uparrow(k) - \xi_\downarrow(k+q)} - \frac{f(\xi_\downarrow(k))}{\omega + i\delta + \xi_\uparrow(k-q) - \xi_\downarrow(k)} \right\}$

$\vec{k} \to \vec{k} - \vec{q}$ in 2nd term:

$$\xi_\uparrow(k-q) - \xi_\downarrow(k) = \varepsilon(k-q) - \varepsilon(k) - \Delta$$

Observe that $\mathrm{Im}\,\Pi^{(0)}_{\uparrow\downarrow} \sim \int (dk)\,(f_\uparrow - f_\downarrow)\,\delta(\omega - \Delta + \varepsilon(k) - \varepsilon(k+q))$

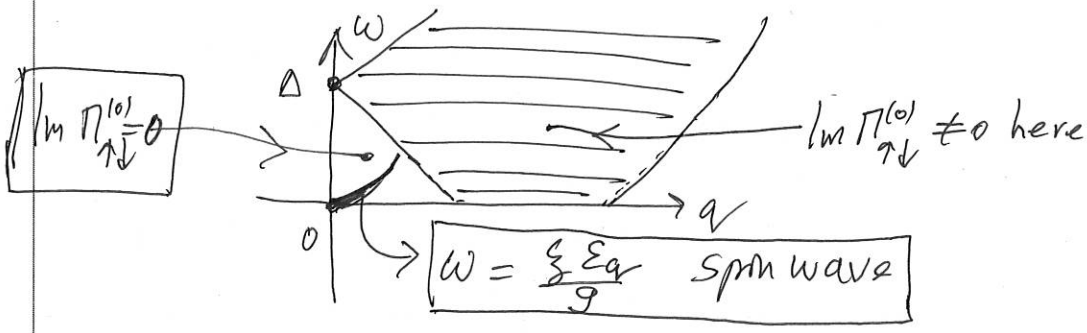
$\Rightarrow$ P-H continuum with $\Delta \neq 0$ can be obtained from that with $\Delta = 0$ by shifting the parabolas along the ω-axis by Δ!



this corner is now free of P-H parts.

$\mathrm{Im}\,\Pi_{\uparrow\downarrow} = 0$ there $\Rightarrow$ collective excitation is possible!

To investigate possible collective mode we need to zoom in on the little $\mathrm{Im}\,\Pi^{(0)}_{\uparrow\downarrow}=0$ region



Clearly we need $\mathrm{Re}(\Pi^{(0)}_{\uparrow\downarrow})$ and are interested in $\omega \ll \Delta$

$\Rightarrow$ let us expand in powers of $\frac{\omega}{\Delta}$, $\frac{\varepsilon(k)}{\Delta}$ ...

let
$$\Pi_1 = \int \frac{d\bar{k}}{(2\pi)^3} \frac{f(\xi_\uparrow(k))}{\omega-\Delta+\frac{k^2-(\bar{k}+\bar{q})^2}{2m}} =$$

$$= -\int \frac{d\bar{k}}{(2\pi)^3} \frac{f(\mu-\frac{\Delta}{2}+\varepsilon(k))}{\Delta-\omega+\frac{q^2+2\bar{k}\cdot\bar{q}}{2m}} = \left(f(E)=\theta(\mu-E) \text{ at } T=0\right)$$

$\varepsilon(q)=\frac{q^2}{2m}$

$$= -\int \frac{d\bar{k}}{(2\pi)^3} \frac{\theta(\mu+\frac{\Delta}{2}-\frac{k^2}{2m})}{\Delta} \frac{1}{1-\frac{\omega-\varepsilon(q)}{\Delta}+\frac{2\vec{k}\cdot\vec{q}}{2m\Delta}}$$

$$= -\int \frac{d\bar{k}}{(2\pi)^3} \frac{\theta(k_F^2-k^2)}{\Delta} \left\{1+\frac{\omega-\varepsilon(q)}{\Delta}-\frac{\vec{k}\cdot\vec{q}}{m\Delta}+\left(\frac{\vec{k}\cdot\vec{q}}{m\Delta}\right)^2\right\}$$

this term is of the order $\frac{q^2}{m^2\Delta^2}$ after k-integration.

We dropped terms like $\left(\frac{\omega-\varepsilon(q)}{\Delta}\right)^2$ because they are of the order $\frac{q^4}{\Delta^2}$.

$$\int \frac{d\vec{k}}{(2\pi)^3} \cdots = \int \frac{dk\, k^2}{(2\pi)^3} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta = \int \frac{dk\, k^2}{(2\pi)^2} \int_{-1}^{1} d(\cos\theta) \ldots$$

Then $\int_{-1}^{1} d(\cos\theta)\ \vec{k}\cdot\vec{q} = \int_{-1}^{1} d(\cos\theta)\ kq\cos\theta = \frac{kq}{2}(\cos\theta)^2\Big|_{-1}^{1} = 0.$

$$\int_{-1}^{1} d(\cos\theta)\ (\vec{k}\cdot\vec{q})^2 = k^2q^2 \int_{-1}^{1} dx\ x^2 = \frac{1}{3}k^2q^2 x^3\Big|_{-1}^{1} = \frac{2}{3}k^2q^2.$$

$$\int_{-1}^{1} d(\cos\theta)\ 1 = 2.$$

$$\Rightarrow \Pi_1 = -\frac{2}{(2\pi)^2}\int_0^{\infty} dk\, k^2\, \frac{\theta(k_\uparrow^2 - k^2)}{\Delta}\left\{1 + \frac{\omega - \varepsilon(q)}{\Delta} + \frac{q^2k^2}{3m^2\Delta^2}\right\}$$

$$q^2 = 2m\,\varepsilon(q)$$

$$\cancel{\Rightarrow \left\{1 + \frac{\omega - \varepsilon(q)}{\Delta} + \frac{2k^2}{3}\frac{\varepsilon(q)}{m\Delta^2}\right\}}$$

$$= \frac{-1}{2\pi^2\Delta}\int_0^{k_\uparrow} dk \left\{k^2\left(1 + \frac{\omega - \varepsilon(q)}{\Delta}\right) + k^4\,\frac{2\,\varepsilon(q)}{3m\Delta^2}\right\}$$

$$= \left(\frac{-1}{2\pi^2}\right)\frac{1}{\Delta}\left\{\frac{k_\uparrow^3}{3}\left(1 + \frac{\omega - \varepsilon_q}{\Delta}\right) + \frac{2\,k_\uparrow^5}{15}\,\frac{\varepsilon_q}{m\Delta^2}\right\}.$$

Similarly $\Pi_2 = \int \frac{d\vec{k}}{(2\pi)^3}\ \frac{f(\xi_\downarrow(k))}{\omega - \Delta + \frac{(\vec{k}-\vec{q})^2 - k^2}{2m}} =$

$$= -\int \frac{d\vec{k}}{(2\pi)^3}\ \frac{\theta(\mu - \frac{\Delta}{2} - \frac{k^2}{2m})}{\Delta}\ \frac{1}{1 - \frac{\omega + \varepsilon_q}{\Delta} + \frac{2\vec{k}\cdot\vec{q}}{2m\Delta}}$$

$$= -\int \frac{d\vec{k}}{(2\pi)^3}\ \frac{\theta(k_\downarrow^2 - k^2)}{\Delta}\left\{1 + \frac{\omega + \varepsilon_q}{\Delta} - \frac{\vec{k}\cdot\vec{q}}{m\Delta} + \left(\frac{\vec{k}\cdot\vec{q}}{m\Delta}\right)^2 + \ldots\right\}$$

$$\Pi_2 = \frac{-2}{(2\pi)^2\Delta}\int_0^{k_\downarrow} dk\, k^2\left(1 + \frac{\omega+\varepsilon_q}{\Delta} + \frac{k^2 q^2}{3m^2\Delta^2}\right)$$

$$= \frac{-1}{2\pi^2\Delta}\left\{\frac{1}{3}k_\downarrow^3\left(1+\frac{\omega+\varepsilon_q}{\Delta}\right) + \frac{1}{5}k_\downarrow^5\,\frac{q^2}{3m^2\Delta^2}\right\}$$

$$q^2 = 2m\varepsilon_q$$

$$\mathrm{Re}\,\Pi_{\uparrow\downarrow} = \Pi_1 - \Pi_2 \cong \frac{-1}{(2\pi^2\Delta)}\left\{\left(1+\frac{\omega}{\Delta}\right)\frac{k_\uparrow^3 - k_\downarrow^3}{3} - \frac{\varepsilon_q}{\Delta}\,\frac{k_\uparrow^3+k_\downarrow^3}{3}\right.$$

$$\left. + \frac{2}{15}\left(k_\uparrow^5 - k_\downarrow^5\right)\frac{\varepsilon_q}{m\Delta^2}\right\} =$$

$$= -\frac{k_\uparrow^3 - k_\downarrow^3}{6\pi^2\Delta}\left\{1 + \frac{\omega}{\Delta} - \frac{k_\uparrow^3+k_\downarrow^3}{k_\uparrow^3-k_\downarrow^3}\,\frac{\varepsilon_q}{\Delta} + \frac{2}{5}\,\frac{k_\uparrow^5-k_\downarrow^5}{k_\uparrow^3-k_\downarrow^3}\,\frac{\varepsilon_q}{m\Delta^2}\right\}$$

$$k_\uparrow^3 - k_\downarrow^3 = \xi\left(k_\uparrow^3 + k_\downarrow^3\right) = 2k_F^3\,\xi$$

$$\Pi'_{\uparrow\downarrow}(q,\omega) = -\frac{k_F^3\xi}{3\pi^2\Delta}\left\{1 + \frac{1}{\Delta}\left[\omega - \frac{2k_F^3}{2k_F^3\xi}\varepsilon_q + \frac{\frac{2}{5}\frac{\varepsilon_q}{m\Delta}(3\pi^2 n)^{5/3}\left[(1+\xi)^{5/3}-(1-\xi)^{5/3}\right]}{2k_F^3\xi}\right]\right.$$

$$k_\uparrow^3 = 6\pi^2 n_\uparrow = 6\pi^2\tfrac{1}{2}n(1+\xi)\ ,\quad k_\downarrow^3 = 6\pi^2\tfrac{1}{2}n(1-\xi)$$

$$k_F^3 = 3\pi^2 n$$

$$2m\Delta = k_\uparrow^2 - k_\downarrow^2 = (3\pi^2 n)^{2/3}\left((1+\xi)^{2/3} - (1-\xi)^{2/3}\right)$$

Thus $$\Pi'_{\uparrow\downarrow} = -\frac{k_F^3\xi}{3\pi^2\Delta}\left\{1 + \frac{1}{\Delta}\left[\omega - \frac{\varepsilon_q}{\xi}\left(1 - \frac{2}{5}\,\frac{(1+\xi)^{5/3}-(1-\xi)^{5/3}}{(1+\xi)^{2/3}-(1-\xi)^{2/3}}\right)\right]\right\}$$

Expand for small $\xi \ll 1$: $$\frac{(1+\xi)^{5/2}-(1-\xi)^{5/2}}{(1+\xi)^{2/3}-(1-\xi)^{2/3}} = \frac{5}{2} - \frac{5\xi^2}{18}$$

$$\Rightarrow \Pi'_{\uparrow\downarrow} = -\frac{k_F^3\xi}{3\pi^2\Delta}\left\{1 + \frac{1}{\Delta}\left(\omega - \frac{\varepsilon_q}{\xi}\cdot\frac{\xi^2}{9}\right)\right\}.$$

The RPA condition: $1 + U\Pi'_{\uparrow\downarrow} = 0$

$$\Rightarrow \quad 1 = \frac{U k_F^3 \xi}{3\pi^2 \Delta}\left\{1 + \left(\omega - \frac{\xi \varepsilon_q}{9}\right)\right\}.$$

Work on the pre-factor: $\dfrac{k_F^3 \xi}{3\pi^2 \Delta}$

$$\Delta = \frac{k_\uparrow^2 - k_\downarrow^2}{2m} = \frac{1}{2m}\underbrace{(3\pi^2)^{2/3} n^{2/3}}_{k_F^2}\left((1+\xi)^{2/3} - (1-\xi)^{2/3}\right)$$

$$= \frac{k_F^2}{2m} \cdot \frac{4}{3}\xi$$

$$\Rightarrow \quad \frac{k_F^3 \xi}{3\pi^2 \frac{k_F^2}{2m} \frac{4\xi}{3}} = \frac{k_F m}{2\pi^2} = \nu$$ just a density of states at the Fermi surface.

$$\Rightarrow \quad 1 = \nu U \left\{ 1 + \omega - \frac{\xi}{9} \frac{q^2}{2m} \right\}.$$

Thus when $\boxed{U\nu = 1}$ is satisfied, there appears a new collective excitation — <u>spin wave</u> with quadratic dispersion

$$\omega = \frac{\xi q^2}{18 m}.$$

$$\Pi^{\parallel(0)}_{\uparrow\downarrow} \sim \frac{\pi}{2}\frac{\omega}{v_F q} \quad \text{for} \quad 0 \le \omega \le v_F q$$

$$\Pi^{\perp(0)}_{\uparrow\downarrow} \sim 1 - \frac{q^2}{12 k_F^2}$$

Approach to ferromagnetic instability: $\Delta = 0 \Rightarrow$ P-H continuum leads to the <u>damping</u> of collective fluctuations.

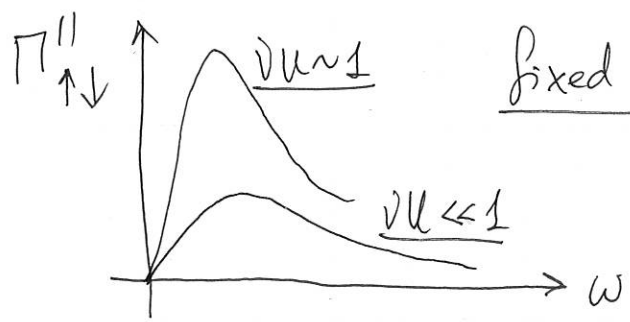
$$1 + U(-\nu)\left[1 - \frac{q^2}{12k_F^2} + i\frac{\pi\omega}{2v_F q}\right] = 1 - \nu U - \nu U \frac{i\pi\omega}{2v_F q}$$

$$\Rightarrow \quad \Pi_{\uparrow\downarrow} = \frac{\Pi^{(0)}_{\uparrow\downarrow}}{1 + U\Pi^{(0)}_{\uparrow\downarrow}} \approx \frac{-\nu}{1 - \nu U - \nu U\left(\frac{i\pi\omega}{2v_F q} - \frac{q^2}{12k_F^2}\right)}$$

Hence, near the instability, when $\nu U \approx 1$

$$\Pi_{\uparrow\downarrow} = \frac{-1/U}{1 - \nu U - \frac{i\pi\omega}{2v_F q} + \frac{q^2}{12k_F^2}}$$

Note that because of factor (i), this dynamics is <u>relaxational</u>

$$\Pi^{\parallel}_{\uparrow\downarrow} \sim \frac{\frac{1}{U}\,\frac{\pi\omega}{2v_F q}}{\left(1 - \nu U + \frac{q^2}{12k_F^2}\right)^2 + \left(\frac{\pi\omega}{2v_F q}\right)^2} = \frac{\frac{\pi\omega}{2v_F q U}}{(1-\nu U)^2 + \left(\frac{\pi\omega}{2v_F q}\right)^2}$$

This function is peaked at $\omega = \frac{2}{\pi} v_F q (1 - \nu U)$